Capstone Proposal: Transparent AI Framework for Identifying and Explaining Claim Denials in Healthcare

Oct 26, 2026

Project information

This proposal is for the Data Science Capstone Project, Winter 2026/2027, and is submitted for approval on 26 Oct 2026.

Field	Detail
Title	Transparent AI Framework for Identifying and Explaining Claim Denials in Healthcare
Course	Capstone Project, Data Science
Term	Winter 2026/2027
Supervisor	Dr. Humera Noor
Team	Dennis Sebastian (59593398), Saran Jayakumaran nair Sasikala (56500625)
Project type	Data, ML and AI: explainable machine learning and NLP
Dataset	California DMHC Independent Medical Review (IMR) Determinations, Trend

Executive summary

We will build a transparent AI framework that predicts whether a denied healthcare service is upheld or overturned on independent review, and explains every prediction in plain language.

Our data is the public Independent Medical Review (IMR) record published by the California Department of Managed Health Care. It covers every IMR decision since 1 January 2001 and is released under a Creative Commons Attribution licence. We compare simple, interpretable baselines with a stronger boosted-tree model, then add an

explanation layer: global and per-case feature attributions, counterfactual "what would change the outcome" examples, and short written reasons grounded in those attributions.

The project delivers a reproducible pipeline, a calibrated model, a fairness audit across age and gender groups, and a demo app. Work runs in six two-week sprints from 26 Oct 2026 and ends with the presentation and pitch on 18 Jan 2027.

Background and problem statement

When a health plan denies, delays or modifies a service, patients rarely learn why, or what would have changed the decision. In California, a patient can ask for an Independent Medical Review when the plan has judged a service not medically necessary, experimental or investigational, or non-emergent or non-urgent. An independent reviewer then decides, and the Department of Managed Health Care publishes the outcomes.

That public record shows which denials were later overturned, but it does not explain why. Machine-learning models can find patterns in it, yet black-box models are hard to trust or contest in a high-stakes domain like health coverage. Explainable AI addresses this by showing which factors drove a prediction.

Problem statement: can we identify, from IMR case attributes, which denials are likely to be upheld or overturned, and explain each prediction transparently enough for a non-expert to understand and challenge it?

Scope note: the dataset holds only denials that reached independent review, not all claim denials. We use the IMR outcome as a measurable proxy for whether a denial was justified, and we state this limit wherever we report results.

Objectives and research questions

The project has five objectives and three research questions.

1. Build a clean, documented, reproducible dataset from the IMR trend file.
2. Train baseline and main models that predict whether the plan's denial is upheld or overturned.
3. Explain predictions globally (what matters overall) and locally (why this case).
4. Turn the explanations into short plain-language reasons and test that they match the model.
5. Audit the models and explanations for gaps across age and gender groups, and package everything in a demo app.

- **RQ1:** Which case attributes, such as diagnosis category, treatment category, review type, age range and year, are most associated with a denial being overturned?
- **RQ2:** How much predictive performance do we give up by using an interpretable model instead of a boosted-tree model?
- **RQ3:** Are the explanations faithful to the model, and are errors similar across age and gender groups?

Dataset

We use the Independent Medical Review (IMR) Determinations, Trend dataset from the California Department of Managed Health Care (DMHC), as shown on the portal on 26 Oct 2026.

Item	Detail
Publisher	California Department of Managed Health Care
Content	All IMR decisions administered by the DMHC since 1 Jan 2001
Coverage	2001 to current
Licence	Creative Commons Attribution (CC BY)
Last updated	8 Oct 2026 on the portal; data files 30 Sep 2026
Update frequency	Irregular
Files	CSV of determinations, PDF data dictionary, ZIP of all resources
Access	CSV download or the portal's CKAN API endpoint

Expected fields. Earlier versions of this dataset describe each case by report year, review type, diagnosis category, treatment category, age range, gender and the determination. We will confirm the exact column names, allowed values and the outcome labels against the PDF data dictionary in Sprint 1, and adjust the target definition if they differ.

Known caveats

- Selection bias: only cases that reached independent review are included.
- Class imbalance is likely between upheld and overturned outcomes, to be measured in Sprint 1.
- Category definitions may have changed over 25 years, so we harmonise them and split train and test by time.

- The file has no claim amounts, plan names or clinical notes we can rely on, which limits what the model can learn.
- The portal updates irregularly, so we pin one downloaded version and record its date and checksum.

Methodology

The method moves from data to explanation in seven steps, using Python with pandas, scikit-learn, a boosted-tree library or InterpretML, SHAP, a counterfactual library such as DiCE, and Streamlit.

1. **Data understanding.** Read the data dictionary, profile the file, and define the target as the plan's denial being upheld versus overturned.
2. **Preparation.** Clean and harmonise categories, group rare values, check for fields that leak the outcome, and split by time so the test set is the most recent years.
3. **Baselines.** Majority-class and logistic regression models set the bar.
4. **Main model.** A gradient-boosted model or an Explainable Boosting Machine, with class weighting, time-aware cross-validation and probability calibration.
5. **Explainability.** Global views (feature importance, SHAP summary) and per-case views (SHAP values, counterfactual what-if cases), cross-checked against the logistic regression coefficients.
6. **Plain-language layer.** Templates turn the top contributing factors into a short written reason. An optional language model may reword the text, but only using the supplied factors, and a template version is always the fallback.
7. **Fairness and evaluation.** Compare error rates and recall across age ranges and genders, test explanation faithfulness, and report limits honestly.

System architecture



system architecture · 7 stages plus audit

Data flows left to right through preparation, then down through modelling, explainability and plain-language reasons into the demo app, while the audit box checks both models and explanations.

Evaluation plan and success metrics

Success means beating the baselines, giving calibrated probabilities, and showing that explanations are faithful and fair across groups. The criteria below are proposed and will be refined with Dr. Noor.

Aspect	Measure	Proposed success criterion
Predictive performance	ROC-AUC, F1 on the minority class, balanced accuracy on a time-based test set	Main model beats the majority-class and logistic regression baselines
Calibration	Brier score, reliability curve	Predicted probabilities track observed outcome rates
Explanation faithfulness	Deletion and insertion tests on top-ranked features; agreement with the interpretable model	Removing top-ranked factors changes the prediction more than removing random ones

Aspect	Measure	Proposed success criterion
Explanation clarity	Short rubric review by peers and supervisor	Reviewers find the written reasons understandable and consistent with the model
Fairness	Recall and error-rate gaps across age ranges and genders	Gaps are measured, reported and discussed, with material gaps flagged
Reproducibility	One-command pipeline with fixed seeds	Results rerun from a clean checkout

Project plan: six two-week sprints

Each sprint ends on the Sunday before a progress report or deliverable is due, so every report summarises a finished sprint.

Sprint	Dates	Goal	Ends with
1	26 Oct to 8 Nov 2026	Foundations and data exploration	Progress report 1 on 9 Nov
2	9 to 22 Nov 2026	Data preparation and baseline models	Progress report 2 on 23 Nov
3	23 Nov to 6 Dec 2026	Main model, tuning and calibration	Progress report 3 on 7 Dec
4	7 to 20 Dec 2026	Explainability, plain-language reasons, fairness audit	Progress report 4 on 21 Dec
5	21 Dec 2026 to 3 Jan 2027	Demo app, evaluation and final report	Final report on 4 Jan 2027
6	4 to 17 Jan 2027	Slides, demo video and pitch	Presentation and pitch on 18 Jan 2027

Task breakdown

Sprint 1: Foundations and data exploration

- 1.1 Submit this proposal and get it approved by Dr. Noor.
- 1.2 Set up the repository, environment, project board and sprint routine.

- 1.3 Download the IMR CSV and data dictionary; keep the raw file read-only and record its date and checksum.
- 1.4 Profile the data: rows per year, missing values, category counts.
- 1.5 Define the target variable, label mapping and class balance.
- 1.6 Write the exploratory notebook: outcomes by year, diagnosis, treatment, age and gender.
- 1.7 Draft the ethics and limitations note.
- 1.8 Progress report 1 and individual LinkedIn post.

Sprint 2: Data preparation and baselines

- 2.1 Clean and harmonise categories; group rare values.
- 2.2 Define the time-based split and run a leakage check.
- 2.3 Build the saved feature pipeline.
- 2.4 Train the majority-class and logistic regression baselines.
- 2.5 Build the metrics harness: ROC-AUC, F1, balanced accuracy, Brier score.
- 2.6 Add unit tests for the data steps.
- 2.7 Progress report 2 and LinkedIn post.

Sprint 3: Main model and tuning

- 3.1 Train the gradient-boosted model or Explainable Boosting Machine.
- 3.2 Handle class imbalance and choose the decision threshold.
- 3.3 Tune hyperparameters with time-aware cross-validation.
- 3.4 Calibrate probabilities and plot reliability curves.
- 3.5 Analyse errors by diagnosis and treatment category.
- 3.6 Produce global explanations and compare them with the logistic coefficients.
- 3.7 Progress report 3 and LinkedIn post.

Sprint 4: Explainability, plain-language reasons, fairness

- 4.1 Generate per-case SHAP explanations.
- 4.2 Generate counterfactual what-if cases.
- 4.3 Build the plain-language template generator.
- 4.4 Add optional language-model rewording, restricted to the supplied factors.
- 4.5 Run faithfulness tests (deletion and insertion).
- 4.6 Run the fairness audit by age range and gender.
- 4.7 Progress report 4 and LinkedIn post.

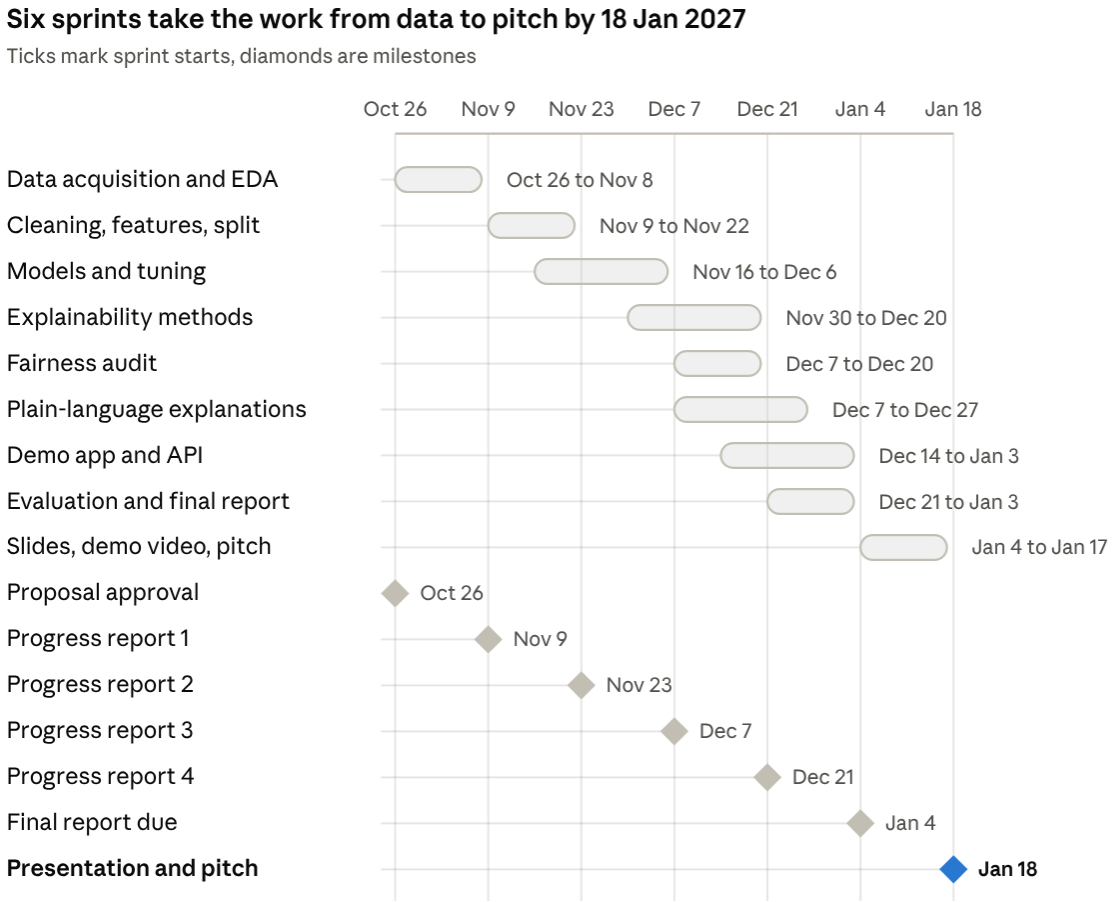
Sprint 5: Demo app, evaluation, final report

- 5.1 Build the Streamlit demo and a small API.
- 5.2 Run the peer rubric review of explanation clarity.
- 5.3 Finalise experiments, tables and figures.
- 5.4 Write the group final report and verify reproducibility.
- 5.5 Submit the final report on 4 Jan 2027.

Sprint 6: Presentation and pitch

- 6.1 Build the slide deck.
- 6.2 Record the 15 to 20 minute presentation video with slides, demo and all members on camera.
- 6.3 Record the elevator pitch video, maximum 2 minutes.
- 6.4 Publish the final LinkedIn post.
- 6.5 Submit links and share access with the supervisor.
- 6.6 Keep buffer time for fixes.

Gantt chart



Gantt chart · 9 workstreams, 7 milestones

Workstreams overlap where one starts while the previous finishes; every sprint boundary lines up with a progress report or deliverable.

Deliverables mapped to course milestones

Date	Milestone	What we submit
26 Oct 2026	Proposal approval	University template or form, this proposal, system diagram, project plan with task breakdown and Gantt chart, LinkedIn post, conversation with AI
9 Nov, 23 Nov, 7 Dec, 21 Dec 2026	Progress reports	Individual LinkedIn post and group status update report each time
4 Jan 2027	Final report	Group final report
18 Jan 2027	Presentation and pitch	Slide deck, 15 to 20 minute presentation video with demo, elevator pitch video of at most 2 minutes, LinkedIn post

Submission links go through Teams or the shared drive, with access given to the supervisor.

Risks, ethics and limitations

Risk	Mitigation
The file has few attributes, so accuracy stays modest	Frame the work as an explanation study; judge success against baselines, not a fixed accuracy
Selection bias: only cases that reached review	State the limit in every result; claim no generalisation to all claim denials
Category definitions change over time	Harmonise categories; split train and test by time
Class imbalance	Class weights, balanced metrics, threshold selection
Language-model text invents reasons	Restrict it to supplied factors, test faithfulness, keep the template version as fallback

Risk	Mitigation
Dataset is updated mid-project	Pin one downloaded version with its date and checksum
Schedule slip or teammate unavailable	Two-week sprints, weekly check-ins, buffer in Sprint 6

Ethics. The data is public and released under CC BY, so we cite the California Department of Managed Health Care as the source. We will confirm in the data dictionary that no direct identifiers are present. The system is a research prototype and must not be used to make real coverage decisions. Fairness results describe this dataset only and are not proof that a decision process is fair.

References

- California Department of Managed Health Care (2026). Independent Medical Review (IMR) Determinations, Trend. CA Open Data Publishing, licence CC BY.
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